

Release # 2 Documentation

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1. Introduction

Overview of the Project

Repository: <https://github.com/vibgetowi/Minicap/tree/main>

Projects board: <https://github.com/orgs/Minicapping-Soen390/projects/3>

The Concordia Campus Guide is a mobile application designed to provide seamless indoor and outdoor navigation across Concordia University campuses. Leveraging React Native and modern development tools, this app aims to enhance the campus experience with features like real-time directions and intuitive navigation for students, faculty, and visitors.

Purpose and Motivation

This project addresses the challenge of navigating large university campuses, where current solutions often lack real-time updates and user-friendly interfaces. By incorporating robust technologies and responsive design, the Concordia Campus Guide ensures accessibility and convenience for its users.

Targeted Users

The application is tailored for:

- **Students and Faculty:** Seeking quick and accurate navigation to classes, offices, and amenities.
- **Visitors:** Requiring easy-to-follow directions for events, tours, and appointments.

The Concordia Campus Guide aspires to be a reliable, efficient, and indispensable tool for all Concordia community members.

2. Requirements

Sprint 1 Backlog

All IDs will have a link to the corresponding user story in ZehHub/GitHub or JIRA. You can also group user stories into EPICs in the table below. This table should be consistent with your Backlog in ZenHub or JIRA at any time during the project. To indicate that a user story has been deleted you can use strikethrough. To indicate a change in the user story points strikethrough the old value and add the new value next to it. Newly added user stories will be highlighted in green color font. **Please link the ID with the corresponding JIRA or ZenHub issue.**

ID	Name	USP	Priority
<u>20</u>	US 1.1 - Support both SGW and Loyola Campus Maps	4	1

ID	Name	USP	Priority
<u>22</u>	US 1.2 - Distinguish Campus Buildings from City Buildings	5	1
<u>23</u>	US 1.3 - Toggle Between SGW and Loyola Maps	3	1
<u>25</u>	US 1.4 - Show Current Location Building	5	2
<u>24</u>	US 1.5 - Show Additional Building Information	3	2
Total		20	

Sprint 2 Backlog

All IDs will have a link to the corresponding user story in ZehHub/GitHub or JIRA. You can also group user stories into EPICs in the table below. This table should be consistent with your Backlog in ZenHub or JIRA at any time during the project. To indicate that a user story has been deleted you can use strikethrough. To indicate a change in the user story points strikethrough the old value and add the new value next to it. Newly added user stories will be highlighted in green color font. **Please link the ID with the corresponding JIRA or ZenHub issue.**

ID	Name	USP	Priority
<u>45*</u>	US 2.1 - Select Start and Destination Building for Outdoor Directions	3	1
<u>46*</u>	US 2.2 - Use Current Location as Start for Outdoor Directions	2	1
<u>47*</u>	US 2.3 - Show Outdoor Directions on Map	3	1
<u>48*</u>	US 2.4 - Directions from SGW to Loyola and Vice Versa	5	2
<u>49*</u>	US 2.5 - Support Multiple Transportation Modes	3	2
<u>50*</u>	US 2.6 - Support for the Concordia Shuttle Service	3	3
Total		19	

Sprint 3 & 4 Backlog

All IDs will have a link to the corresponding user story in ZehHub/GitHub or JIRA. You can also group user stories into EPICs in the table below. This table should be consistent with your Backlog in ZenHub or JIRA at any time during the project. To indicate that a user story has been

deleted you can use strikethrough. To indicate a change in the user story points strikethrough the old value and add the new value next to it. Newly added user stories will be highlighted in green color font. **Please link the ID with the corresponding JIRA or ZenHub issue.**

ID	Name	USP	Priority
<u>51*</u>	US 3.1 - View Class Schedule and Classroom Locations	4	1
<u>52*</u>	US 3.2 - Generate Directions to Next Class	2	2
<u>53*</u>	US 4.1 - Locate Rooms on a Specific Floor	6	1
<u>54*</u>	US 4.2 - Show Shortest Indoor Path Directions	3	2
<u>55*</u>	US 4.3 - Show Indoor Directions for Users with Disabilities	3	2
<u>56*</u>	US 4.4 - Highlight Indoor Points of Interest	2	3
<u>57*</u>	US 4.5 - Show Directions Between Rooms on Different Floors	5	3
<u>58*</u>	US 4.6 - Indoor Directions from SGW to Loyola and Vice Versa	2	3
Total		27	

3. Release Planning

This will be repeated 3 times for the 3 Sprints of each Release. So, after each release there should be 3 Sprint summaries in the document.

Sprint 1 Summary

Sprint 1 primarily focused on setting up the basic app structure and defining team workflows. However, the team faced challenges with the late addition of new members and establishing initial processes, which delayed the actual development work. As a result, the first sprint was not representative of the project's expected progress. While the MVP 0: Basic App Setup and Structure was initiated, most planned user stories, including campus mapping features, were pushed to Sprint 2.

Story ID		Planned USP	Status
<u>13</u>	MVP 0: Basic App Setup and Structure	3	STARTED IN SPRINT 1

Story ID		Planned USP	Status
			WILL CONTINUE IN SPRINT 2
<u>20*</u>	US 1.1 - Support both SGW and Loyola Campus Maps	4	PLANNED FOR SPRINT 2
<u>22*</u>	US 1.2 - Distinguish Campus Buildings from City Buildings	5	PLANNED FOR SPRINT 2
<u>23*</u>	US 1.3 - Toggle Between SGW and Loyola Maps	3	PLANNED FOR SPRINT 2
<u>25*</u>	US 1.4 - Show Current Location Building	5	PLANNED FOR SPRINT 2
<u>24*</u>	US 1.5 - Show Additional Building Information	3	PLANNED FOR SPRINT 2
<u>45*</u>	US 2.1 - Select Start and Destination Building for Outdoor Directions	2	
<u>46*</u>	US 2.2 - Use Current Location as Start for Outdoor Directions	2	
<u>47*</u>	US 2.3 - Show Outdoor Directions on Map	6	

Story ID		Planned USP	Status
<u>48*</u>	US 2.4 - Directions from SGW to Loyola and Vice Versa	3	
<u>49*</u>	US 2.5 - Support Multiple Transportation Modes	3	
<u>50*</u>	US 2.6 - Support for the Concordia Shuttle Service	2	
<u>51*</u>	US 3.1 - View Class Schedule and Classroom Locations	4	
<u>52*</u>	US 3.2 - Generate Directions to Next Class	2	
<u>53*</u>	US 4.1 - Locate Rooms on a Specific Floor	6	
<u>54*</u>	US 4.2 - Show Shortest Indoor Path Directions	3	
<u>55*</u>	US 4.3 - Show Indoor Directions for Users with Disabilities	3	
<u>56*</u>	US 4.4 - Highlight Indoor Points of Interest	2	

Story ID		Planned USP	Status
<u>57</u> *	US 4.5 - Show Directions Between Rooms on Different Floors	5	
<u>58</u> *	US 4.6 - Indoor Directions from SGW to Loyola and Vice Versa	2	
<u>59</u> *	US 5.1 - Show Nearest Outdoor Points of Interest		
<u>60</u> *	US 5.2 - Show Directions to a Selected Outdoor Point of Interest		
<u>61</u> *	US 6.1 - Create an Optimized Task Plan		
<u>62</u> *	US 6.2 - Directions for Optimized Plan		
Total		23	X

An * would be a good way to show stories added during the sprint. And a strikethrough would be good to denote a deleted story. Use anything you want, just make sure this information is clear.

Project velocity after 1 sprint: X

Sprint 2 Summary

Sprint 2 marked significant progress, with the team successfully implementing key mapping functionalities. The app now supports navigation for both SGW and Loyola campuses (**US 1.1**), differentiates campus buildings from city structures (**US 1.2**), and allows users to toggle between maps (**US 1.3**). Additionally, work began on showing the user's current location (**US 1.4**) and providing additional building information (**US 1.5**), though both features require further

refinement in Sprint 3. With the campus mapping foundation in place, the team is now preparing to implement outdoor navigation and routing features in the next sprint.

Story ID		Planned USP	Status
<u>13</u>	MVP 0: Basic App Setup and Structure	3	STARTED IN SPRINT 1 WILL CONTINUE IN SPRINT 3
<u>20*</u>	US 1.1 - Support both SGW and Loyola Campus Maps	4	FINISHED IN SPRINT 2
<u>22*</u>	US 1.2 - Distinguish Campus Buildings from City Buildings	5	FINISHED IN SPRINT 2
<u>23*</u>	US 1.3 - Toggle Between SGW and Loyola Maps	3	FINISHED IN SPRINT 2
<u>25*</u>	US 1.4 - Show Current Location Building	5	STARTED IN SPRINT 2 WILL CONTINUE IN SPRINT 3 (2 tasks remaining)
<u>24*</u>	US 1.5 - Show Additional Building Information	3	STARTED IN SPRINT 2 WILL CONTINUE IN SPRINT 3 (1 task remaining)
<u>45*</u>	US 2.1 - Select Start and Destination Building for Outdoor Directions	3	PLANNED FOR SPRINT 3
<u>46*</u>	US 2.2 - Use Current Location as	2	PLANNED FOR SPRINT 3

Story ID		Planned USP	Status
	Start for Outdoor Directions		
<u>47*</u>	US 2.3 - Show Outdoor Directions on Map	3	PLANNED FOR SPRINT 3
<u>48*</u>	US 2.4 - Directions from SGW to Loyola and Vice Versa	5	PLANNED FOR SPRINT 3
<u>49*</u>	US 2.5 - Support Multiple Transportation Modes	3	PLANNED FOR SPRINT 3
<u>50*</u>	US 2.6 - Support for the Concordia Shuttle Service	3	PLANNED FOR SPRINT 3
<u>51*</u>	US 3.1 - View Class Schedule and Classroom Locations		
<u>52*</u>	US 3.2 - Generate Directions to Next Class		
<u>53*</u>	US 4.1 - Locate Rooms on a Specific Floor		
<u>54*</u>	US 4.2 - Show Shortest Indoor Path Directions		
<u>55*</u>	US 4.3 - Show Indoor Directions for		

Story ID		Planned USP	Status
	Users with Disabilities		
<u>56</u> *	US 4.4 - Highlight Indoor Points of Interest		
<u>57</u> *	US 4.5 - Show Directions Between Rooms on Different Floors		
<u>58</u> *	US 4.6 - Indoor Directions from SGW to Loyola and Vice Versa		
<u>59</u> *	US 5.1 - Show Nearest Outdoor Points of Interest		
<u>60</u> *	US 5.2 - Show Directions to a Selected Outdoor Point of Interest		
<u>61</u> *	US 6.1 - Create an Optimized Task Plan		
<u>62</u> *	US 6.2 - Directions for Optimized Plan		
Total		42	X

An * would be a good way to show stories added during the sprint. And a strikethrough would be good to denote a deleted story. Use anything you want, just make sure this information is clear.

Project velocity after 1 sprint: X

Sprint 3 & 4 Summary

Sprint 3 and 4 focused on enhancing outdoor navigation and route planning, bringing significant improvements to the app. The team successfully completed showing outdoor directions on the map (**US 2.3**) and providing directions between SGW and Loyola (**US 2.4**).

Additionally, work began on supporting multiple transportation modes (**US 2.5**) and integrating Concordia's shuttle service (**US 2.6**), both of which will continue into Sprint 5. Progress was made on viewing class schedules and classroom locations (**US 3.1**), laying the groundwork for generating directions to the next class (**US 3.2**) in Sprint 5.

Sprint 5 will focus on indoor navigation (**US 4.1 - US 4.6**), expanding upon existing mapping capabilities.

Story ID		Planned USP	Status
<u>13</u>	MVP 0: Basic App Setup and Structure	3	STARTED IN SPRINT 1 WILL CONTINUE IN SPRINT 4
<u>20*</u>	US 1.1 - Support both SGW and Loyola Campus Maps	4	FINISHED IN SPRINT 2
<u>22*</u>	US 1.2 - Distinguish Campus Buildings from City Buildings	5	FINISHED IN SPRINT 2
<u>23*</u>	US 1.3 - Toggle Between SGW and Loyola Maps	3	FINISHED IN SPRINT 2
<u>25*</u>	US 1.4 - Show Current Location Building	5	FINISHED IN SPRINT 3

Story ID		Planned USP	Status
<u>24</u> *	US 1.5 - Show Additional Building Information	3	FINISHED IN SPRINT 2
<u>45</u> *	US 2.1 - Select Start and Destination Building for Outdoor Directions	3	FINISHED IN SPRINT 3
<u>46</u> *	US 2.2 - Use Current Location as Start for Outdoor Directions	2	FINISHED IN SPRINT 3
<u>47</u> *	US 2.3 - Show Outdoor Directions on Map	3	FINISHED IN SPRINT 4
<u>48</u> *	US 2.4 - Directions from SGW to Loyola and Vice Versa	5	FINISHED IN SPRINT 4
<u>49</u> *	US 2.5 - Support Multiple Transportation Modes	3	STARTED IN SPRINT 4 WILL CONTINUE IN SPRINT 5 (2 tasks remaining)
<u>50</u> *	US 2.6 - Support for the Concordia Shuttle Service	3	STARTED IN SPRINT 4 WILL CONTINUE IN SPRINT 5 (2 tasks remaining)
<u>51</u> *	US 3.1 - View Class Schedule and Classroom Locations	4	STARTED IN SPRINT 4 WILL CONTINUE IN SPRINT 5 (3 tasks remaining)

Story ID		Planned USP	Status
<u>52</u>*	US 3.2 - Generate Directions to Next Class	2	PLANNED FOR SPRINT 5
<u>53</u>*	US 4.1 - Locate Rooms on a Specific Floor	6	PLANNED FOR SPRINT 5
<u>54</u>*	US 4.2 - Show Shortest Indoor Path Directions	3	PLANNED FOR SPRINT 5
<u>55</u>*	US 4.3 - Show Indoor Directions for Users with Disabilities	3	PLANNED FOR SPRINT 5
<u>56</u>*	US 4.4 - Highlight Indoor Points of Interest	2	PLANNED FOR SPRINT 5
<u>57</u>*	US 4.5 - Show Directions Between Rooms on Different Floors	5	PLANNED FOR SPRINT 5
<u>58</u>*	US 4.6 - Indoor Directions from SGW to Loyola and Vice Versa	2	PLANNED FOR SPRINT 5
<u>59</u>*	US 5.1 - Show Nearest Outdoor Points of Interest		
<u>60</u>*	US 5.2 - Show Directions to a Selected Outdoor Point of Interest		

Story ID		Planned USP	Status
61*	US 6.1 - Create an Optimized Task Plan		
62*	US 6.2 - Directions for Optimized Plan		
Total		69	X

An * would be a good way to show stories added during the sprint. And a strikethrough would be good to denote a deleted story. Use anything you want, just make sure this information is clear.

Project velocity after 1 sprint: X

Burndown Chart

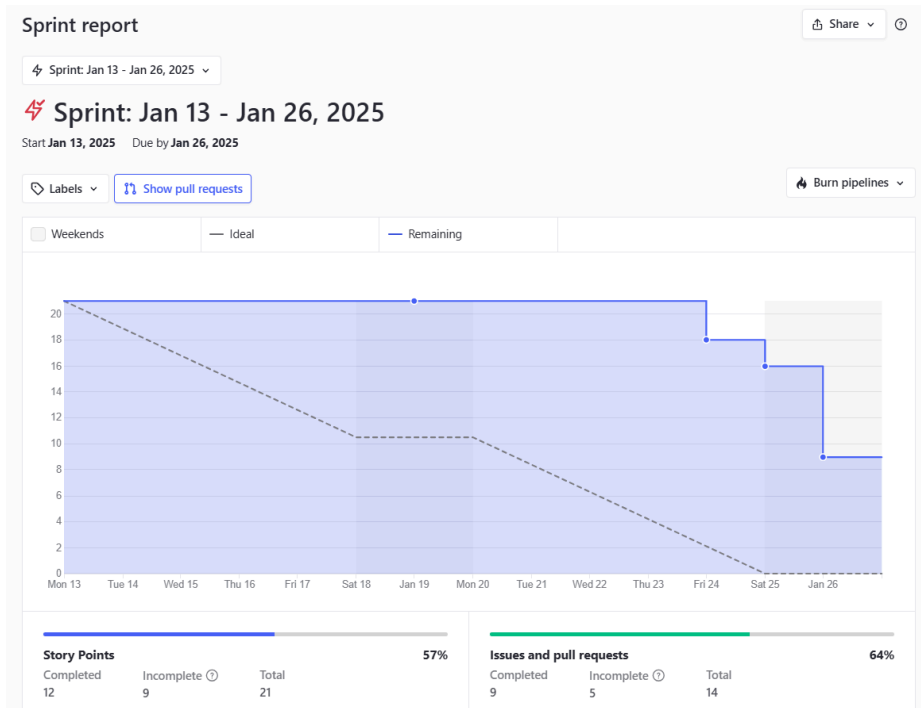
Due to problems with Zenhub, burndown charts are not accurate

<https://app.zenhub.com/workspaces/soen-390-678d66b2d0a94300315d4867/reports/burndown?milestoneId=Z2lkOi8vcmludC8yOTM3MzM2>

Sprint 1 Retrospective

<https://github.com/AsifAliKhan2001/Minicap/blob/main/Meeting-Notes/release-1/2025-01-27%20-%20Biweekly.md>

The chaos of setting up and recruiting new members was a significant issue as well as figuring out our processes. Due to course complications work started one week late, the first sprint should not be representative

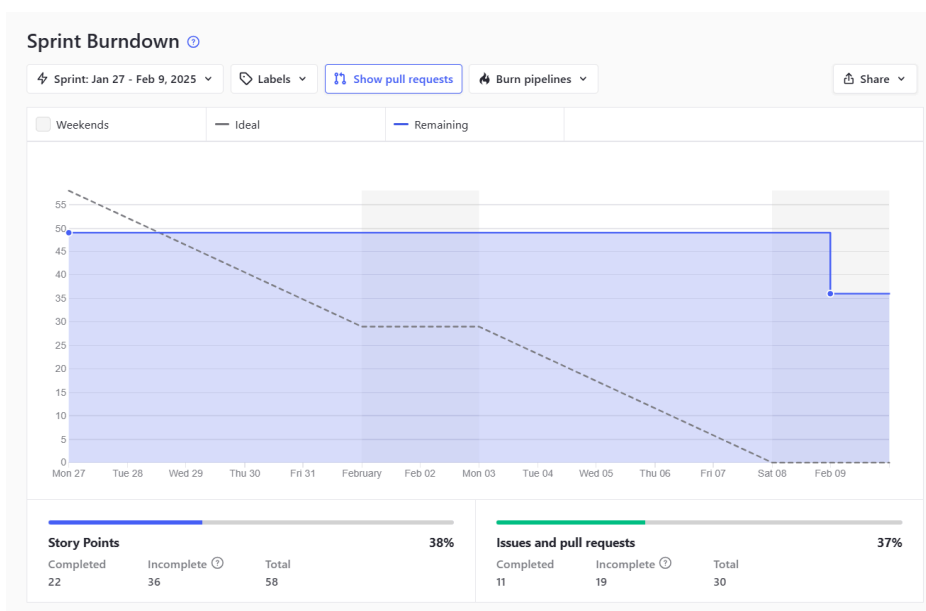


<https://app.zenhub.com/workspaces/soen-390-678d66b2d0a94300315d4867/reports/burndown?milestoneId=Z2lkOi8vcmludC8yOTM3MzM3>

Sprint 2 Retrospective

<https://github.com/vibgetowi/Minicap/blob/main/Meeting-Notes/release-1/2025-02-12%20Sprint%20Retrospective.md>

Zenhub being down created issues with task tracking, this has now been fixed, we should be set for sprint 3



Sprint 3

<https://github.com/vibgetowi/Minicap/blob/dev/Meeting-Notes/release-2/2025-03-03%20Sprint%20Retrospective.md>

The midterm period has reintroduced chaos in our availabilities

For both sprint 3 and 4, transitioning to a github organisation for deployment means unfortunately the burndown charts will have to wait as it broke the code

You can check out the progress of forking the charting tool here:

<https://github.com/vibgetowi/github-projects-burndown-chart>

Sprint 4

[pending retro link as retros are done after the sprint]

4. Architecture

4.1 Overview of the architecture

- For all diagrams please refer to the wiki <https://github.com/AsifAliKhan2001/Minicap/wiki/System-Diagrams> they change often so no images here
- Conventions for diagrams are to capture cardinalities with relationships, being tech agnostic means we also don't define lists nor FKs directly. Some inheritances are omitted (like to audit) for visual clarity but a note was provided

4.6 Design Patterns

We use MVVM architectural pattern for separation of concerns and maintainability, see our detailed writeup on our wiki <https://github.com/AsifAliKhan2001/Minicap/wiki/MVVM-guide>

More design patterns will be implemented once we get to the Indoor Navigation

4.7 External libraries

Technology Stack

Please refer to our README <https://github.com/vibgetowi/Minicap/blob/dev/README.md>

And wiki page on backend decisions:

<https://github.com/Minicapping-Soen390/Minicap/wiki/Backend-Stack>

5. User Interface Design

5.1 Personas

<https://github.com/AsifAliKhan2001/Minicap/wiki/UI-UX-Documentation#personas>

5.3 Mockups

<https://github.com/AsifAliKhan2001/Minicap/wiki/UI-UX-Documentation#wireframes->

6. Testing Plan and Report

<https://github.com/Minicapping-Soen390/Minicap/wiki/QA-Test-Results>

6.1 Unit Testing

<https://github.com/Minicapping-Soen390/Minicap/wiki/QA-Test-Results#261-unit-testing>

6.2 Test code coverage

<https://github.com/Minicapping-Soen390/Minicap/wiki/QA-Test-Results#262-test-code-coverage>

6.3 Acceptance Testing

Link to Acceptance Testing with TA approval.

https://docs.google.com/spreadsheets/d/18Wtc-ROJasvmf_37tXjRKmPG577l3O61qfTMKf7J0ac/edit?gid=0#gid=0

Github:

<https://github.com/Minicapping-Soen390/Minicap/wiki/QA-Test-Results#263-acceptance-testing>

6.4 System Tests

<https://github.com/Minicapping-Soen390/Minicap/wiki/QA-Test-Results#264-system-tests>

6.5 SonarQube Report

Sonarqubes report have 2 week data gap due to SonarQubes responsible being MIA

<https://github.com/Minicapping-Soen390/Minicap/wiki/QA-Test-Results#266-sonarqube-report>

6.6 Usability Testing

<https://github.com/Minicapping-Soen390/Minicap/wiki/QA-Test-Results#265-usability-testing>

7. Defect Tracking Report

For each release you will include 3 links with the bug reports for each Sprint.
In GitHub you can simply add the following suffix to your project url
issues?q=**label**%3Abug+**is**%3Aclosed+**milestone**%3A"Sprint+3"

where **label** indicates the label you use for bug-related issues (typically bug)
is indicates the status of the bug (open or closed)
milestone indicates the Sprint

This is the link to view bugs

<https://github.com/Minicapping-Soen390/Minicap/issues?q=is%3Aissue%20label%3Abug%20>

Release 2 bugs:

- <https://github.com/Minicapping-Soen390/Minicap/issues/202>
- <https://github.com/Minicapping-Soen390/Minicap/issues/179>
- <https://github.com/Minicapping-Soen390/Minicap/issues/166>

8. Code review report

1) <https://github.com/vibgetowi/Minicap/pull/163>

Lots of engaging discussions and proposed functionalities and refactorings that promote reusability and better UX

2) <https://github.com/Minicapping-Soen390/Minicap/pull/216>

A great PR because we have a complex feature and the developer mostly respect our style conventions and MVVM, with a slight error which was caught by the SCRUM team and communicated to the developer

3) <https://github.com/Minicapping-Soen390/Minicap/pull/201>

Not the proudest PR but it is an important record of why it is important to keep up communication with dev subteams .

4) <https://github.com/Minicapping-Soen390/Minicap/pull/168>

A thorough PR and also showing how we uphold standards by ensuring to not merge anything not following our style guides.

5) <https://github.com/Minicapping-Soen390/Minicap/pull/165>

This one is the first PR that applied the new much improved format, we are very proud of it as the step up in quality is significant. Also discussions of future refactoring strategies as we noticed one file was becoming too complex

9. Refactoring report

<https://github.com/vibgetowi/Minicap/wiki/Refactoring-Tracker>

We have an ongoing refactoring effort for CampusMap.tsx as it has become too complex (complexity of ~140)

10. Quality Measurements

At the end of each Sprint record the value for each of the following metrics: size, duplication, documentation, complexity, coupling, cohesion, technical debt, security vulnerabilities, and code convention violations.

You should take actions in each Sprint to improve these metrics. List the commits and pull requests in which technical debt, security vulnerabilities, and code convention violations have been addressed. Link to your Refactoring report actions taken to improve complexity, duplication, coupling, cohesion, readability and other design/code quality metrics.

Finally, create a chart showing the evolution of each metric over the Sprints.

Everything can be found here:

<https://github.com/Minicapping-Soen390/Minicap/wiki/QA-Test-Results>

11. Professional ethics and accountability

There may be accessibility concerns which we address in our persona section of our wiki, linked above

Appendix

So far, ai is not being used